higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CONCRETE STRUCTURES
NQF LEVEL 3

21 NOVEMBER 2019

This marking guideline consists of 6 pages.

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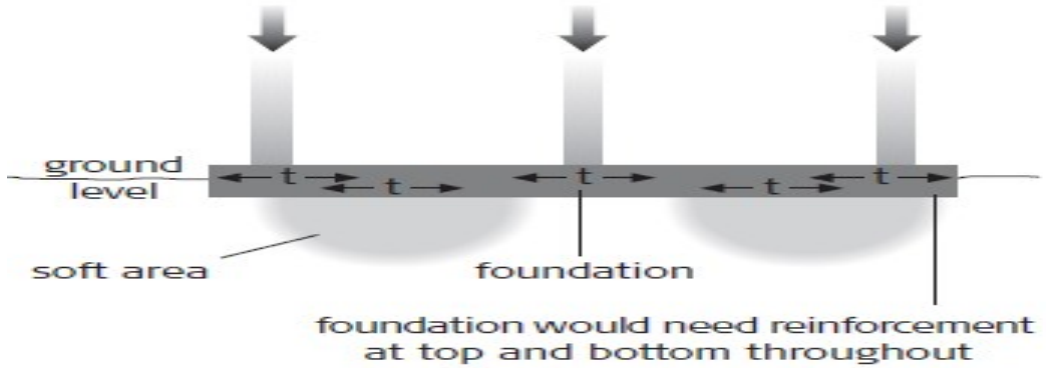
QUESTION 1: THE THEORY AND USE OF REINFORCED CONCRETE

1.1

1.1.1	True
1.1.2	True
1.1.3	False
1.1.4	True
1.1.5	False
1.1.6	False
1.1.7	True
1.1.8	False
1.1.9	False
1.1.10	False

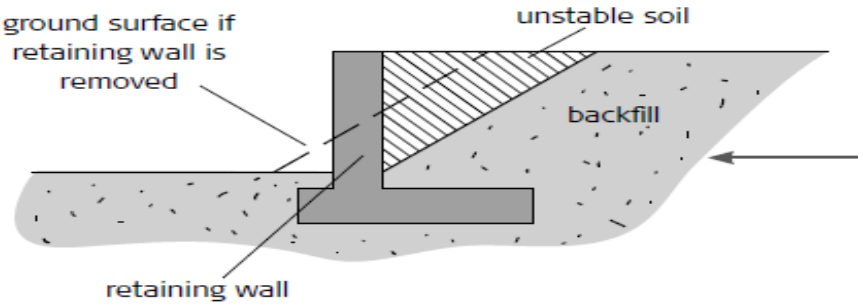
(10 × 1) (10)

1.2



Mark allocation
Raft foundation
Labelling: 3
Correctness: 2
Total = 5 (5)

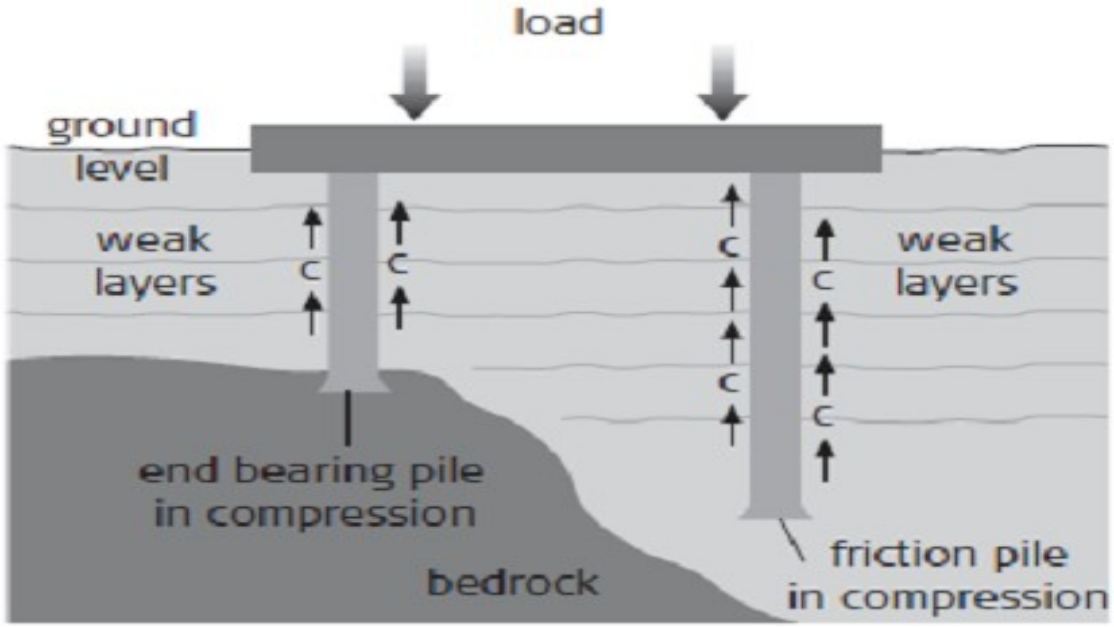
1.3



A CANTILEVER RETAINING WALL

Mark allocation
Correctness:
Title: 1
Labels: 3
Total = 5 (5)

1.4



PILE FOUNDATION

Mark allocation
Correctness: 1
Title: 1
Labels: 3
Total = 5 (5)

[25]

QUESTION 2: REINFORCED CONCRETE IN CONSTRUCTION

2.1
2.1.1 B
2.1.2 C
2.1.3 D
2.1.4 E
2.1.5 A

(5 × 1) (5)

2.2
- Sheet metal drawing✓: The sheets are shaped by drawing them over round or curved objects.✓
- Bar, wire and tube drawing✓ : The metal is drawn through a very strong metal object called a die. It is thick when it enters the die, and long and thin when it exits.✓

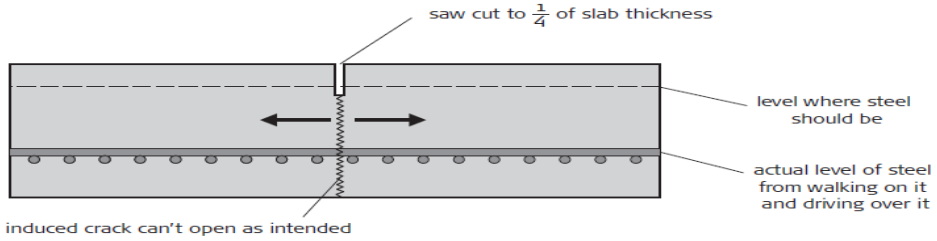
(4)

2.3
R✓ – Refers to plain round bars of strength of 250 MPA✓
Y✓ – Refers to high-yield deformation steel bars of strength 450✓
Z✓ – Refers to types of steel not covered by R or Y✓
SM✓ – Refers to standard mesh (high tensile wire of strength 485 MPA)✓
DM✓ – Refers to design mesh (high tensile wire of strength 485 MPA)✓

(Any 4 × 2) (8)

2.4 8 mm,✓ 10 mm,✓ 12 mm,✓ 16 mm,✓ 20 mm,✓ 25 mm,✓ 32 mm✓
(Any 3 × 1) (3)

2.5



Mark allocation
Correctness: 1
Labels: 4
Total = 5 (5)

[25]

QUESTION 3: REINFORCING DOCUMENTATION

3.1 3.1.1 A list of all reinforcement needed for a structure showing the bar marks, cutting length, number of bars needed and the different sizes and shapes

3.1.2 A number or a letter that comes before other numbers or letters

3.1.3 A small allowance difference to specification measurement

3.1.4 A set of rules people agree to use

3.1.5 A number or a letter that comes after other numbers or letters
(5 × 1) (5)

3.2 3.2.1 Normally refer to either reinforcing bars or a concrete edge that can be seen

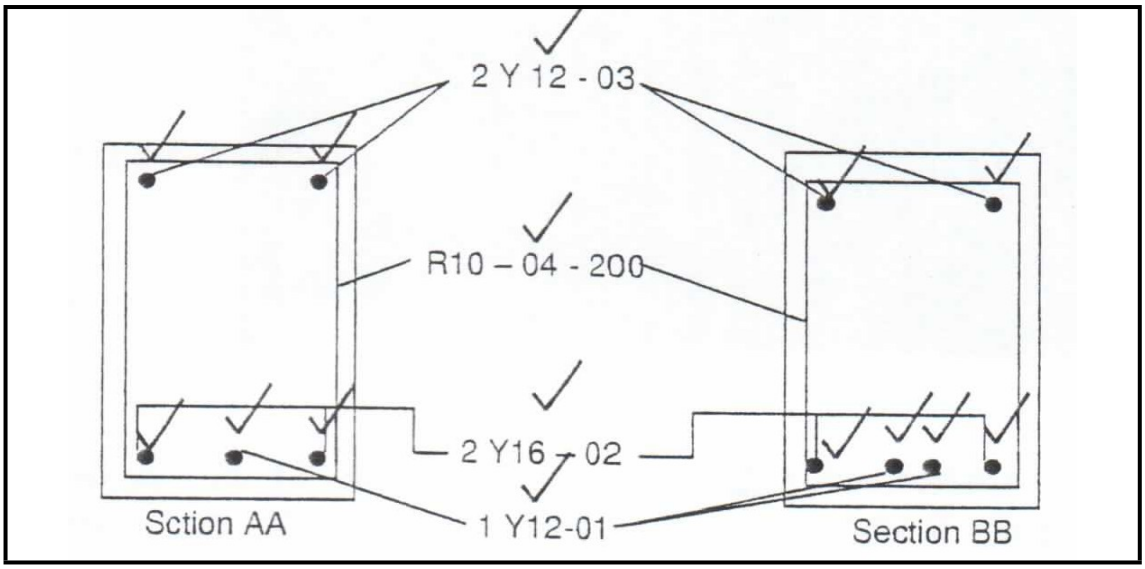
3.2.2 Normally refer to dimension lines

3.2.3 Normally refer to reinforcing bars details on the drawings that cannot be seen

3.2.4 Normally refer to centre lines

3.2.5 Normally refer to view through a section of an object
(5 × 1) (5)

3.3



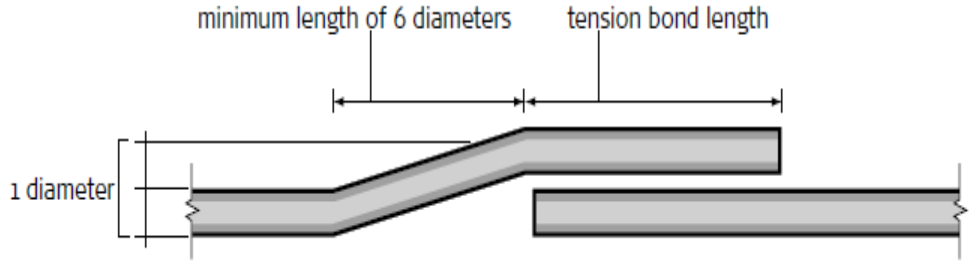
(15)
[25]

QUESTION 4: FABRICATION OF REINFORCING STEEL

4.1

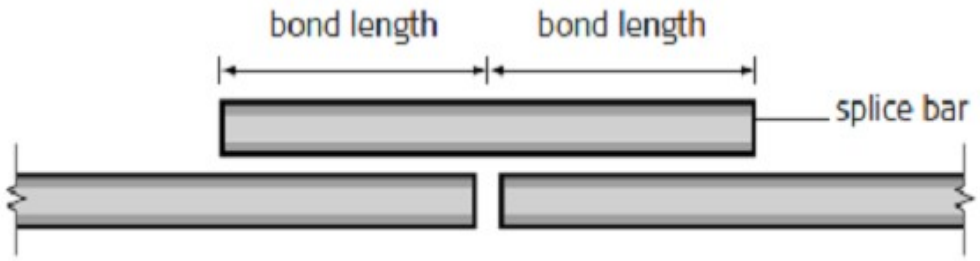
- Identify cut steel required for bending.
- Take needed steel from the cut-steel racks.
- Use data from schedules for bending the steel.
- Confirm bending lengths, shape, thickness and type of steel.
- Enter data from schedule into bending machine.
- Feed steel into bending machine and bend steel according to specification.
- Bundle and tag bars.
- Move bent steel into loading racks. (Any 8 × 1) (8)

4.2



SPLICING BY CRANKING ONE BAR

Mark allocation
Correctness: 1
Labels: 2,5
Total = 3.5



A SPLICE BAR

Mark allocation
Correctness: 1
Labels: 2,5
Total = 3,5

(7)

4.3
- While workers walk on it, it can move.
- While concrete is being placed, the steel can move.
- During compacting or vibration of the cast concrete the cage or mat can move. (3)

4.4 Step 1: Stirrups are placed first – all bunched up at one end.
Step 2: Steel is then put in at the bottom and the top.
Step 3: Marks are then made on the bars where the stirrups must go.
Step 4: The stirrups are moved in place and tying begins.
Step 5: Finally the clips/spacers are placed and secured. (5)

4.5 4.5.1 A reinforced cage is the finished reinforcement for columns and beams.

4.5.2 A mat is the finished reinforcement for slabs or bases.

(2 ×1) (2)
[25]

TOTAL: 100